

SYSTEMATIC SEARCH FOR SINGULARITIES IN 3D EULER-VOIGT FLOWS

SYSTEMATIC SEARCH FOR SINGULARITIES IN 3D EULER-VOIGT FLOWS

By NOAH BENSLEER, B.Sc.

A Thesis Submitted to the Department of Computational Science & Engineering
and the School of Graduate Studies of McMaster University
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Acknowledgements

An expression of thanks for assistance given by the Supervisor of research and by others should be either set forth on a separate page or incorporated into the Preface (if there is one). These and all subsequent preliminary pages listed under (f), (g) and (h) must be numbered in lower case Roman numerals, i.e., 'iv', 'v', 'vi' etc.

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1 Introduction

1.1 3D Euler-Voigt Equations

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1.2 Initial Conditions

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1.3 Finite Time Blow-up

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2 The Optimization Problem

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2.1 The Adjoint Method

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2.2 Gâteaux (Directional) Differential

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2.3 Riemannian Conjugate Gradient Method

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3 Numerical Methods

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3.1 Spatial Discretization & Spectral Methods

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3.2 Time Stepping

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3.3 Kappa Test Validation

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4 Results

4.1 Results for $\alpha = \text{value 1}$

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4.2 Results for $\alpha = \text{value 2}$

TEXT

4.3 Results for $\alpha = \text{value 3}$

TEXT

5 Conclusion

TEXT

6 Appendix

TEXT example citation [1]

References

[1] John Nash. Non-cooperative games. *Annals of Mathematics*, 54(2):286–295, 1951.